

the owner's username, passwords, high scores, etc. Since every user will have specific pre-defined text, we will save it to the database and retrieve it when needed.

6. **Texting component** - This enables us to perform various SMS related functionalities. We will need this to send automatic messages
7. **Phone call component** - This component will detect all call related activity like identifying an incoming call or dialling a number. In our case, we need to track the incoming calls to get the number of the caller, hence we are including this in our project.

Given below are the components that we need for this application, which we will drag from the left hand side palette on to the designer.

	Component name	Purpose	Location
1	Horizontal Arrangement	To arrange the child components horizontally	Palette--> Layout--> Horizontal Arrangement
2	Label	To display a label	Palette-->User Interface-->Label
3	Text Box	To receive input from the user	Palette-->User Interface-->Text Box
4	Button	To confirm the user choice	Palette-->User Interface-->Button
5	Tiny Data-base	To store the phone number locally	Palette--> Data-base--> Tinydb
6	Texting	To control the SMS related functionality	Palette-->Social-->Texting
7	Phone call	To control the call related functionality	Palette-->Social-->Phone call

Here are the things that you already know, to begin with.

1. Drag and drop the components mentioned above to the viewer.
2. Visible components can be seen while the non-visible components will be located beneath the viewer under the tag *Non-visible*.
3. Label and text box have to be put within the Horizontal Arrangement to keep them aligned.
4. If you have dragged and placed everything, the layout will look something like what's shown in Figure 1.

Now make the necessary property changes for the label and button components. At this stage we do not have any media usage so we will not upload any image or sound to our application.

Your graphical user interface is ready. Figure 2 shows exactly how the application will look like after the installation.



Figure 1: Getting started



Figure 2: A GUI application

We will now head to the blocks editor to define its behaviour so let's discuss the actual functionality expected from our application.

1. Within the app, the user should be able to store the automated text which will be delivered to the mobile number attached with each incoming SMS or call. The user will type in the text box.
2. When any incoming message is received, the application should be able to extract the number attached to the incoming message and send back the automated message. The text will be pre-saved in the database.
3. When any incoming call is detected and it has not been answered, only then should the application be able to extract the number attached to the incoming call and send back the automated message, because there is no point in sending the SMS if the call has been answered by the user.

So let's move on and add these three functions, using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the *Properties* pane to switch between the designer and block editor.

Block Editor

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop it on the viewer. Arrange the blocks in the same way shown in Figure 3. I will explain what each block does and how it is called.

1. The block comes into action as soon as the user presses the button. Its functionality is to save whatever text the user has typed into the text box into the local phone database called *TinyDB*. To store anything into the

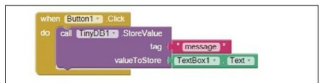


Figure 3: Arrangement of blocks in block editor